A non-periodic sequence

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The sequence (a_n) is defined by $a_1 = 2$ and $a_{n+1} = \left[\frac{3}{2}a_n\right]$ for every $n \geq 1$. Show that the sequence (b_n) defined by $b_n = (-1)^{a_n}$ is not eventually periodic.

(Recall that $[x]$ denotes the integer part of x.)